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MULTI-STAGE DECISION METHOD FOR PRODUCTION OF NEW TRANSLATION EXAMPLES

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ABSTRACT

It is very important for Machine Translation Method using Inductive Learning with Genetic Algorithms(GA-ILMT) which we have already developed that the correct rate of new translation examples is high rate and there are more translation examples produced. We propose the method which makes new translation examples produced by one-point crossover the correct of rate and increase. In this paper, we describe our method and some experiments.

KEY WORDS

genetic algorithms, inductive learning, machine translation, multi-stage decision method

1 INTRODUCTION

Recently, as computer networks are progressing, the chance to contact with many other languages is increasing. To understand language except native language, a machine translation system is effective. However, the machine translation system has problems on the correct translation rate and the quality of translation results. The machine translation system mainly uses analytical method. It can not analyze exceptional sentences due to the use of limited gr ammatical rules. To resolve this problem, we introduce a learning method to machine translation system[1]. If this system could get enough translation examples, it could translate the exceptional sentences. However this system has the problem that it requires many translation examples.

To resolve this problem, we previously proposed a method of Machine Translation using Inductive Learning with Genetic Algorithms(GA-ILMT), and this method has been evaluated[2]. In this method, Genetic Algorithms are applied to translation examples for production of new translation examples, and translation rules are extracted from them. In the process for production of new translation examples, two translation examples which have the common parts are selected. And the common parts are decided as crossover positions. Next, one-point crossovers for these two translation examples are performed. Therefore even though there are a small number of translation examples, many new translation examples could be produced by Genetic Algorithms. However the correct rate of new translation examples are not enough high.

In this paper, we propose Multi-Stage Decision Method for Crossover Position for Production of New Translation Examples to resolve this problem. And We confirm that our purposed method is effective in production of new translation examples.

2 PROCESS IN THIS METHOD

The system based on our proposed method decides corresponding relationships between words of source sentences of two translation examples and between words of target sentences of them. And using these results, corresponding relationships between words of source sentences and target sentences of them are decided.

In GA-ILMT, the crossover positions are only common character strings between two translation examples. However, in our proposed method, the words decided the corresponding relationships are decided as crossover position. Then one-point crossovers are performed, and produced two new translation examples. Figure 1 shows the outline of the system based on our proposed method.

2.1 PROCESSES BETWEEN SOURCES SENTENCES AND BETWEEN TARGET SENTENCES.

In the decisions of corresponding relationships for words between source sentences and between target sentences, the corresponding relationships are decided as follows:

(1) the corresponding relationships between words with the same character strings and position in sentences.

(2) the corresponding relationships between words with the same character strings.

(3) source sentences:

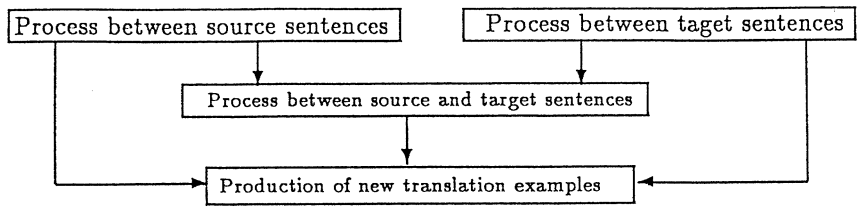


Figure 1: OUTLINE OF THE SYSTEM

Classifcation	Name	Number	Manufacture	Stage	Poupose
Thesaurus	WordNet	126,516	Princeton University	3	Irreglar conjugation
				5	Superclass consept (English)
	bunrui-goi-hyo	32,600	*Kokuritu-kokugo-kenkyujo*	5	Superclass consept (Japanese)
Morpheme-dictionary	Oxford Advanced Learner's Dictionary of Current English	70,646	University of London	3	Regular conjuction
English-Japanese dictionary	*Eiwa・waei-densaku-jisyo*	55,000	*Seiji Kubo* (*Gijyutu-hyouronsya*)	4	Translation word
List of translation words	ONE WORLD English Cource 1 NEW EDITION (Word list)	389	*Kiyoshi Hasegawa* etc	4	Translation word
List of translation words	NEW HORIZON English Cource 1 (Word list)	397	*Akira oota* etc	4	Translation word
Tools of Morpheme-anaryze	Brill Tagger	1,429,865	Eric Brill	7	A part of speach (English)
	Morphological anallysis method using inductive learning	11,113	*Kenji araki*	3	Reading of word
				7	A part of speach (Japanese)

Table 1: DICTIONARY AND TOOLS USED IN PROCESSES

the corresponding relationships between words with same the original form of word in source sentences.

target sentences:

the corresponding relationships between words with same reading in tag et sentences.

(4) the corresponding relationships between same translated words.

(5) the corresponding relationships between words with the same superclass concept.

(6) the corresponding relationships between a word which put between the decided words.

(7) the corresponding relationships between word with a same part of speech.

Table 1 shows the dictionary and tools used in these processes. The words decided before the stages is not processed in the stage. And the value of similarity between two sentences is calculated by equation (1).

$$VS = \frac{100.0}{n} \times (2.0 \times N1 + 1.8 \times N2 + 1.6 \times N3 + 1.2 \times N4 + 0.8. \times N5 + 0.4 \times N6 + 0.2 \times N7 - 0.5 \times N8) \quad (1)$$

,where:

n:the total number of words of the two sentences.

VS:the value of similarity between two sentences.

N1,N2,....,N7 are the number of words of (1) (2)....(7).

N8 is the number of words which have the corresponding relationships.

2.2 PROCESS BETWEEN SOURCE AND TARGET SENTENCES

This process decides the corresponding relationships between source sentence and target sentence. We use English-Japanese dictionary[3] in this process. This system looks for the words of target sentence in this dictionary, then the word of source sentence with the corresponding relationships are searched.

2.3 PRODUCTION OF NEW TRANSLATION EXAMPLES

In production of new translation examples, the words of corresponding relationships in these sentences are crossover position as common part. Then we apply one-point crossovers to a pair of two translation examples. The condition to decide word of crossover position is as follows:

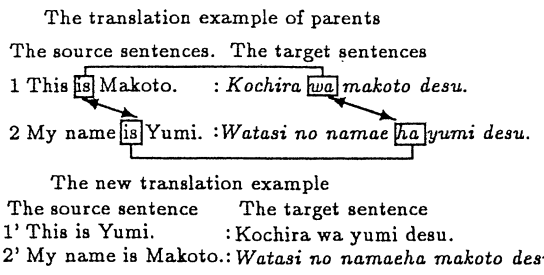


Figure 2: THE EXAMPLE OF THE CORESSPONDING RELATIONSHIPS AND NEW TRANSLATION

1 the word of each sentence has the corresponding relationship between two source sentences or between two target sentences.

2 the word of each sentence has the corresponding relationship between source sentence and target sentence.

If the word satisfies those conditions, the word is defined as crossover position. As shown in Figure 2, the corresponding relationships of words between source sentences and between target sentences. And the corresponding relationships of "is" and "*ha*" are decided in the process between source and target sentences. Because these words satisfy these conditions, these words are regard as common part and these words are crossover position. Then one-point crossover is performed in the source sentences and the target sentences. Then new two translation examples are produced. All words fulfilling conditions treat as crossover position.

3 EXPERIMENTS FOR PERFORMANCE EVALUATION

3.1 STANDARD OF EVALUATION

The new translation examples are classified into two categories:

(1)the new source sentence and the new target sentence are correct.

(2) the new source is not correct, or the new target sentence is not correct.

We experiment the correct rate and the number of the new translation examples.

3.2 PROCEOVRE OF EXPERIMENTS

We performed the production experiment of new translation examples using translation examples between English and Japanese by our proposed method. And 1,746 translation examples were used as data, of which 948 examples were taken from a textbook[4] for first grade junior high school students, and 800 examples from another textbook[5]. We process each of the same lesson of both these

Table 2: TOTAL AND THE CORRECT RATE OF NEW TRANSLATIONS

The name of method	correct	mistake	rate%
GA-ILMT	2146	5332	12.3
The our method	2835	4715	37.5

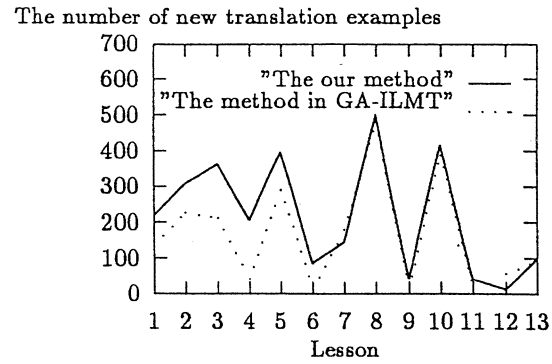


Figure 3: THE NUMBER OF CORRECT TRANSLATION

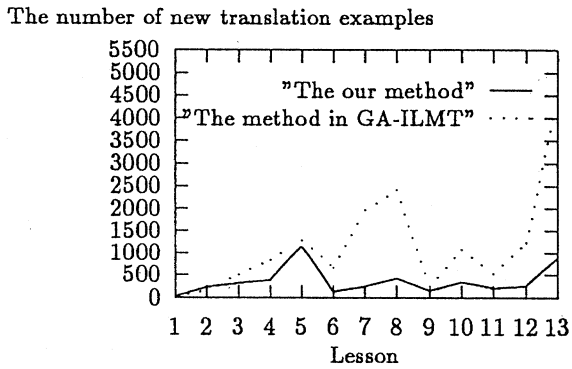


Figure 4: THE NUMBER OF ERRONEOUS NEW TRANSLATIONS

textbooks on the system. And in the decision of corresponding relationships and the produce of new translation examples ,all translation examples in each of the same lesson are object of processing. The one-point crossover is applied to translation examples over 25point of the value of similarity between two sentences calculated by equation (1). The experiment using only former method of GA-ILMT is performed too.

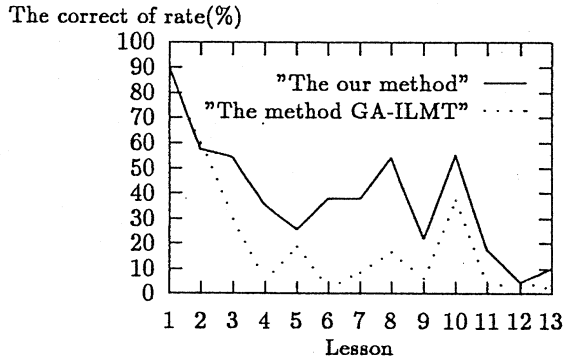


Figure 5: THE CORRECT RATE OF NEW TRANSLATIONS

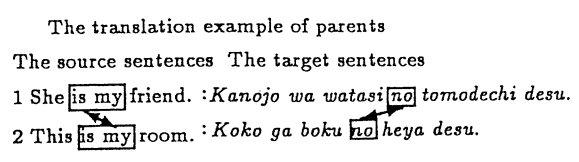


Figure 6: THE EXAMPLE OF NEW TRANSLATION IN GA-ILMT

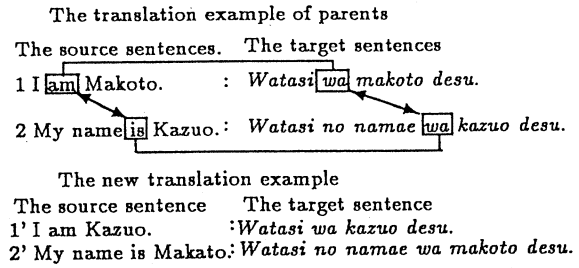


Figure 7: THE EXAMPLE OF NEW TRANSLATION IN THIS METHOD

3.3 RESULTS OF EXPERIMENT

The rate of new translation examples increased from 12.3% to 37.5%. And the number of effective new translation examples increased from 2,146 to 2,835. Table 1 and Figure 3,4,5 show result of experiments.

4 DISSCUTION

We restrict the value of similarity between two sentences, because the pair of similar translation example is production object. Therefore, the number of correct new translation examples and the correct rate increased. The second, in Figure.6, the crossover positions are only common character strings between two translation examples without the corresponding relationships between source and target sources in GA-ILMT. However, as shown Figure.7, in our proposed method, we use many informations which are the various corresponding relationships of words between sentences to decide on crossover position. Therefore the crossover position has high certainty. And crossover positions are not only wards of common character strings in GA-ILMT, however crossover positions are words decided by various informations in our method. Therefore new translation examples which could not have ever product are produced by our proposed method. Therefore we think that new translation examples increase. In the experiments, the results show that our purposed method is effective in production of new translation examples.

5 CONCLUSION

In this paper, we proposed Multi-stage decision method for production of new translation examples. By using this method, the correct rate of new translation examples increased by 28.4%, and the number of correct new translation examples increased by 689. Therefore, we consider that our proposed method is effective in production of new translation examples. In future, we plan to apply our purposed method to GA-ILMT.

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